Wabi Demeke Diriba, Ph.D.

Computational Modeling • Physics-Informed AI/ML • Design Optimization

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Summary

PhD in Mechanical Engineering with over seven years of experience accelerating engineering design at the intersection of computational physics and AI/ML. Deep expertise in **physics-informed ML**, **ML-based design optimization**, and **Bayesian optimization** to build highly accurate **surrogate models** under data-scarce conditions. Possess a strong foundation in scientific workflows integrating physics-based data generation via **multiphysics FEM simulation** with ML model training and rigorous validation. Author of eight peer-reviewed publications including first-author contributions in top-tier computational and energy journals. Most recently engineered **multi-agent LLM systems** to autonomously orchestrate atomistic computational modeling pipelines.

Education

Ph.D., Mechanical Engineering **2020 – 2024**
Korea Advanced Institute of Science and Technology (KAIST), South Korea *Advisor: Prof. Seunghwa Ryu*
Dissertation: Data-Driven Design for Energy Harvesting Applications

M.S., Mechanical Engineering **2018 – 2020**
KAIST, South Korea *Advisor: Prof. Seunghwa Ryu*
Thesis: ANN-based optimization of segmented thermoelectric generators

B.S., Mechanical Engineering **2014 – 2018**
KAIST, South Korea

Experience

Distinguished Postdoctoral Fellow **Nov 2025 – Present**
University of Wyoming *Laramie, WY, USA*

- Architected an autonomous **multi-agent LLM framework** that plans, executes, and self-corrects atomistic transport simulations (DFTB) on **HPC clusters**, reducing expert setup time from days to minutes (manuscript in preparation).
- Automated **end-to-end data generation** from DFTB outputs to electronic transport solvers and developing crystal **graph convolutional neural networks** for rapid prediction of electronic transport properties in metal-organic frameworks (MOFs) (work in progress).

Applied AI Engineer **May 2025 – Nov 2025**
Enable Fusion (startup) *Daejeon, South Korea*

- Built **RAG-based AI tools** to streamline engineering consultation workflows for fusion energy customers.
- Directed development of **physics-informed neural networks** for precision manufacturing of tokamak structural components, translating customer requirements into deliverable ML solutions.

Postdoctoral Researcher **Sep 2024 – May 2025**
KAIST, Laboratory for AI-powered Modeling in Mechanics, Materials, and Manufacturing *Daejeon, South Korea*

- Led device-level fatigue-life analysis of thermoelectric power generators using **multiphysics FEM** sub-modeling, linking field-relevant thermal cycling loads to structural reliability metrics.
- Mentored Ph.D. researchers and co-developed **physics-informed ML** methods that embed governing equations directly into learning, enabling **label-free inverse identification** of temperature-dependent material properties with **uncertainty quantification** in small-data regimes; published in *npj Computational Materials* (2025) and featured in *Phys.org*.
- Led a comprehensive review of thermoelectric power generators covering material discovery, structural optimization, and service-life stability; published in *Renewable and Sustainable Energy Reviews*.

Graduate Research Assistant **Sep 2018 – Aug 2024**
KAIST, Laboratory for AI-powered Modeling in Mechanics, Materials, and Manufacturing *Daejeon, South Korea*

- Developed **ML-driven design optimization** of segmented thermoelectric generators using **deep learning surrogates** trained on high-fidelity simulation data, with **active–transfer learning** to efficiently explore design spaces of $\sim 10^{14}$ candidates infeasible with direct simulation.
- Applied **Bayesian optimization** to design gradient-index phononic crystals for **vibration energy harvesting**, maximizing elastic energy focusing with few expensive simulation evaluations.
- Developed an **ML-based genetic algorithm** framework coupling deep learning surrogates with evolutionary search for free-form design of plate phononic crystals, achieving wide bandgaps for vibration attenuation.
- Automated **multiphysics FEM workflows** via **Python/MATLAB scripting of COMSOL**, enabling rapid, large-scale simulation data generation for ML surrogate modeling.
- Delivered results across several **government-funded projects**, including quantifying the effect of interfacial resistance on thermoelectric device performance and structural analysis for fatigue-life prediction.

Visiting Researcher **Feb 2023 – Mar 2023**
Korea Electrotechnology Research Institute (KERI) *Changwon, South Korea*

- Developed **AI-driven optimization** of thermoelectric material properties, accelerating material screening for thermal energy harvesting devices in collaboration with KERI scientists.

Research Intern **Jul 2017 – Aug 2017**
Korea Institute of Energy Research (KIER) *Daejeon, South Korea*

- Built a GUI coupling economic analysis with combustion process simulations.

Research Intern **Jan 2017 – Jun 2017**
KAIST, Laboratory for AI-powered Modeling in Mechanics, Materials, and Manufacturing *Daejeon, South Korea*

- Automated generation of graphene bicrystal configurations for molecular dynamics simulations via MATLAB.

Core Skills

Scientific ML: physics-informed machine learning (PIML), deep learning surrogate models, Bayesian optimization, active & transfer learning, genetic/multi-objective optimization

Simulation: COMSOL Multiphysics (expert), coupled thermal–electric–structural FEM, transient & steady-state heat transfer, fatigue & reliability analysis, VASP, DFTB

ML Engineering: PyTorch, TensorFlow, scikit-learn, automated simulation-to-training data pipelines, HPC/Slurm job orchestration, RAG & multi-agent LLM frameworks

Programming: Python, MATLAB, simulation macro automation, Linux

Publications

- **Demeke, W.**†, Lee, J.†, Lee, I.†, Ryu, B.*, Ryu, S.* (2026). A comprehensive review of thermoelectric power generators: material discovery, structural optimization, and service-life stability. *Renewable and Sustainable Energy Reviews* (accepted).
- **Demeke, W.**, Lee, S., Jeon, W., Ryu, S. (2026). Enhancing elastic energy focusing in multimode strain regions via Bayesian optimization of gradient-index phononic crystals. *Int. J. Solids & Structures*, 113871.
- Moon, H., Lee, S., **Demeke, W.***, Ryu, B., Ryu, S. (2025). Physics-informed neural operators for generalizable, label-free inference of temperature-dependent thermoelectric properties. *npj Computational Materials*, 11, 272.
- Ryu, B., Oh, S., **Demeke, W.**, et al. (2025). Wiedemann–Franz law and thermoelectric inequalities. *J. Applied Physics*, 137(11), 115001.
- Moon, H., Cho, H., **Demeke, W.**, Ryu, B., Ryu, S. (2025). Thermal conductivity estimation with uncertainty quantification using Bayesian physics-informed neural networks. *arXiv:2510.16723*.
- **Demeke, W.**, Ryu, B., Ryu, S. (2024). Machine learning-based optimization of segmented thermoelectric power generators using temperature-dependent performance properties. *Applied Energy*, 355, 122216.

- **Demeke, W.**, et al. (2023). Free-form shape optimization of plate phononic crystals with wide bandgaps via deep neural networks. *Extreme Mechanics Letters*, 65, 102098.
- **Demeke, W.**, et al. (2022). Neural network-assisted optimization of segmented thermoelectric generators using active learning and genetic optimization. *Energy Reports*, 8, 6633–6644.
- Jung, J., **Demeke, W.**, et al. (2021). Micromechanics-based prediction of thermoelectric properties of anisotropic composites and porous media. *Int. J. Thermal Sciences*, 165, 106918.

†Equal contribution *Corresponding author

Selected Conference Presentations

- **Demeke, W.**, et al. (2026). From Text to Transport: An LLM-based multi-agent pipeline for automated DFTB transport calculations in MOFs. *Gordon Research Conference on Computational Materials Science and Engineering*, Newry, ME, USA. (Poster)
- **Demeke, W.**, Ryu, B., Ryu, S. (2023). ML-based optimization of segmented thermoelectric power generators. *MRS Spring Meeting*, Materials Research Society. (Oral)
- Five additional oral/poster presentations at KSME Spring/Autumn Conferences (2020–2024) on Bayesian leg-shape design, DNN-based bandgap optimization, and active transfer learning.

Funded Research Projects

- **Lead contributor**, Physical fatigue-life model at the thermoelectric device level (N06240077), 2024–2025.
- **Contributor**, Electrical–thermal circuit modeling of short-leg thermoelectric devices (N05220008), 2022.
- **Contributor**, Multiscale modeling of thermoelectric phenomena in composite materials (N05200011), 2020.

Honors & Awards

- KAIST Ph.D. Fellowship (competitive international-student scholarship & stipend), 2020–2024.
- KAIST Full Scholarship for B.S., M.S., and Ph.D., awarded for academic performance, 2014–2024.

References

Available upon request.